Communication

Experimental Investigation and Thermodynamic Modeling for the Mg-Nd-Sr System

GUANGLONG XU, YUWEN CUI, LIJUN ZENG, XIAOMA TAO, LIBIN LIU, and ZHANPENG JIN

Microstructure examination, phase analysis, and DSC measurement on the equilibrated ternary alloys not only enabled exploration of the phase equilibria relations of the Mg-Nd-Sr ternary but also thermodynamic assessment of the ternary system in the context of the CALPHAD approach. Thermodynamic modeling, further coupled with first principle calculations, was thereafter used to predict a panoramic phase diagram for the Mg-Nd-Sr ternary system at the Mg-rich corner.

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Magnesium alloys with rare-earth (RE) elements have proved effective in enhancement of creep resistance and elevated temperature performance.[1] Among these alloys, the Mg-Al-(Ca/Sr) and Mg-Al-RE alloys surpass most of commercial die casting Mg alloys in tensile strengths at elevated temperature including AZ91D,[2] arising from solid-solution strengthening and/or formation of precipitates when the formers are alloyed with RE and/or Sr.[3] The CALPHAD (CALculation of PHAse Diagram) approach,[4] appropriately assisted with key experimentation and first principle calculations enables to exploit the phase diagram of the Mg-Al-Ca-Sr-RE multicomponent system for designing high-performance RE-containing Mg alloys. Zhong *et al.*,[5,6] Janz *et al.*[7,8] and Aljarrah *et al.*[9,10] published the thermodynamic assessments for the quaternary Mg-Al-Ca-Sr system. Nevertheless, there is neither phase equilibrium information nor thermodynamic description for the Mg-RE-Sr systems. It is the objectives of this work to determine the phase equilibria relations of the Mg-Nd-Sr ternary system by studying the equilibrated Mg-rich ternary alloys and by using differential scanning calorimeter analysis, and upon the determined data to develop the thermodynamic descriptions of the Nd-Sr binary and Mg-Nd-Sr ternary systems in terms of the CALPHAD approach.

The Mg-Nd-Sr ternary alloys were prepared from the binary master alloys. The weighed samples were wrapped in tantalum foils and then sealed in pure iron tubes under argon atmosphere. The tubes were heated up and held at 1173 K (900 °C) for 4 hours for melting and then quenched into ice water. Each tube was turned upside down every 45 minutes to promote a homogeneous composition. The samples were kept in the tubes for the heat treatment at 673 K (400 °C) for 720 hours, followed by a water quench. The actual compositions of the annealed samples were analyzed by the chemical titration technique, and the microstructure and phase present/composition were examined by powder X-ray diffraction (XRD) using a Dmax-2500 VBX diffractometer and by scanning electron microscopy/electron microprobe analysis (SEM/EPMA) using a JEOL-JXA-8230. Small pieces were cut from the samples for the heat flux DSC measurement using a TA-SDTQ600 at a heating/cooling rate of 10 K min^{-1}.

The results obtained from the chemical titration, XRD, SEM/EPMA, and DSC investigations to all the alloy samples are summarized in Table I. Although a noticeable difference in phase compositions were observed in different samples, it scarcely alters the determined phase relationship. In fact, the identical compositions of the three phases at equilibrium resulting from different samples were impossible to obtain even by repeated melting and long-term annealing of the alloys particularly due to the inherent volatility of Mg and the high melting temperature of Nd as well as the lack of fluidity in sealed tube; fortunately, a difference of 1 to 2 at. pct is considered to be within the experimental uncertainty. It is due to this fact that the detected solubility of binary compounds was given with a lower weight in subsequent thermodynamic assessment. Figure 1(a) shows the SEM image of Alloy #1 (Mg-2.52 at. pct Nd-3.31 at. pct Sr) after annealing at 673 K (400 °C) for 720 hours while the XRD pattern is given in Figure 1(b). The EPMA and XRD results reveal that the dark phase is (Mg) having little Nd and Sr solubility, the gray one is $Mg_{17}Sr_2$ with about 2 at. pct Sr substituted by Nd, and the light one is $Mg_{41}Nd_5$ that exhibits noticeable substitution of 4 at. pct Sr (also supported by Alloy 2#) for Nd while the Mg content remains about 89 at. pct. It should be stressed that, although the long- term annealing failed to essentially alter the solidification morphology presented in many of the samples, the compositional homogeneity of all the phases has been well evidenced by the extensive EPMA measurements in all detectable area of view of the samples. Together with the DSC measurements, the examination of microstructure is also helpful in determining the invariant reactions and transformation pathways. For instance, as shown in Figure 1(a), Alloy #1 embraces a microstructure consisting of the primary (Mg) and the [(Mg) + $Mg_{17}Sr_2$] and [(Mg) + $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$] eutectics that concludes a ternary eutectic reaction Liq → (Mg) + $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$. Because the binary eutectics has remarkably

GUANGLONG XU, Ph.D. Student, is with the School of Materials Science and Engineering, Central South University, Changsha 410083, P.R. China, and also with Computational Alloy Design Group, IMDEA Materials Institute, 28906 Getafe, Madrid, Spain, and also with the Facultad de Informática, Universidad Politécnica de Madrid, 28660 Boadilla del Monte, Madrid, Spain. YUWEN CUI, Staff Researcher, is with the Computational Alloy Design Group, IMDEA Materials Institute. LIJUN ZENG, Ph.D. Student, LIBIN LIU, and ZHANPENG JIN, Professors, are with the School of Materials Science and Engineering, Central South University. Contact e-mail: lbliu@csu.edu.cn XIAOMA TAO, Professor, is with the College of Physical Science and Technology, Guangxi University, Nanning 530004, P.R. China.

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Table I. Phases Present and Compositions Determined by XRD, SEM/EDS, and DSC Measurements for the Annealed Mg-Nd-Sr Ternary Alloys

No.	Chemical Composition (at. pct)			BSE Phases Analysis	Microprobe Analysis		Primary Crystal	XRD	DSC Effects	Microstructure and Invariant Reaction
	Mg	Sr	Nd		Sr	Nd				
1	94.17	3.31	2.52	(Mg)	0.01	0.03	(Mg)	(Mg)	839.03	(Mg) + [(Mg) + $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$]$_e$
				$Mg_{17}Sr_2$	6.98	2.57		$Mg_{17}Sr_2$		Liq → (Mg) + $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$
				$Mg_{41}Nd_5$	3.38	6.41		$Mg_{41}Nd_5$		(839)
2	90.33	7.57	2.10	(Mg)	0	0.04	$Mg_{17}Sr_2$	(Mg)	839.16	$Mg_{17}Sr_2$ +
				$Mg_{17}Sr_2$	8.14	2.14		$Mg_{17}Sr_2$	856.41	[(Mg) + $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$]$_e$
				$Mg_{41}Nd_5$	4.48	6.47		$Mg_{41}Nd_5$		
3	90.4	1.88	7.72	(Mg)	0.02	0.06	(Mg)	(Mg)	854	(Mg) + [(Mg) + $Mg_{41}Nd_5$]$_e$
				$Mg_{41}Nd_5$	2.12	8.50		$Mg_{41}Nd_5$		
4	86.83	2.07	11.10	Mg_3Nd	0.25	23.88	$Mg_{41}Nd_5$	Mg_3Nd	n/a	Mg_3Nd + [$Mg_{41}Nd_5$ + Mg_3Nd]$_e$
				$Mg_{41}Nd_5$	2.41	8.98		$Mg_{41}Nd_5$		
5	89.33	7.92	2.75	$Mg_{17}Sr_2$	9.20	1.74	$Mg_{17}Sr_2$	$Mg_{17}Sr_2$	860.18	($Mg_{17}Sr_2$) + $Mg_{41}Nd_5$
				$Mg_{41}Nd_5$	4.44	6.85		$Mg_{41}Nd_5$		(856)
6	86.66	12.85	0.49	$Mg_{17}Sr_2$	9.45	1.52	$Mg_{17}Sr_2$	$Mg_{17}Sr_2$	846.27*	($Mg_{17}Sr_2$) + $Mg_{38}Nd_9$
				$Mg_{38}Sr_9$	18.36	1.13		$Mg_{38}Sr_9$	856.65	
								$Mg_{41}Nd_{5\ (trace)}$	865.15	
7	71.38	24.52	4.10	$Mg_{23}Sr_6$	19.97	1.74	Mg_2Sr	$Mg_{23}Sr_6$	n/a	Mg_2Sr + [Mg_3Nd + $Mg_{23}Sr_6$]$_e$
				Mg_2Sr	31.22	1.47		Mg_2Sr		
				Mg_3Nd	0.16	23.54		Mg_3Nd		
8	83.26	12.23	4.51	$Mg_{23}Sr_6$	20.02	1.59	$Mg_{23}Sr_6$	$Mg_{23}Sr_6$	852.08	
				Mg_3Nd	0.11	23.76		Mg_3Nd		
				$Mg_{41}Nd_5$	3.31	8.05		$Mg_{41}Nd_5$		

The ternary invariant reactions in the table are deduced from the microstructure observation and DSC measurements.
*The value with star is the diffuse signal.

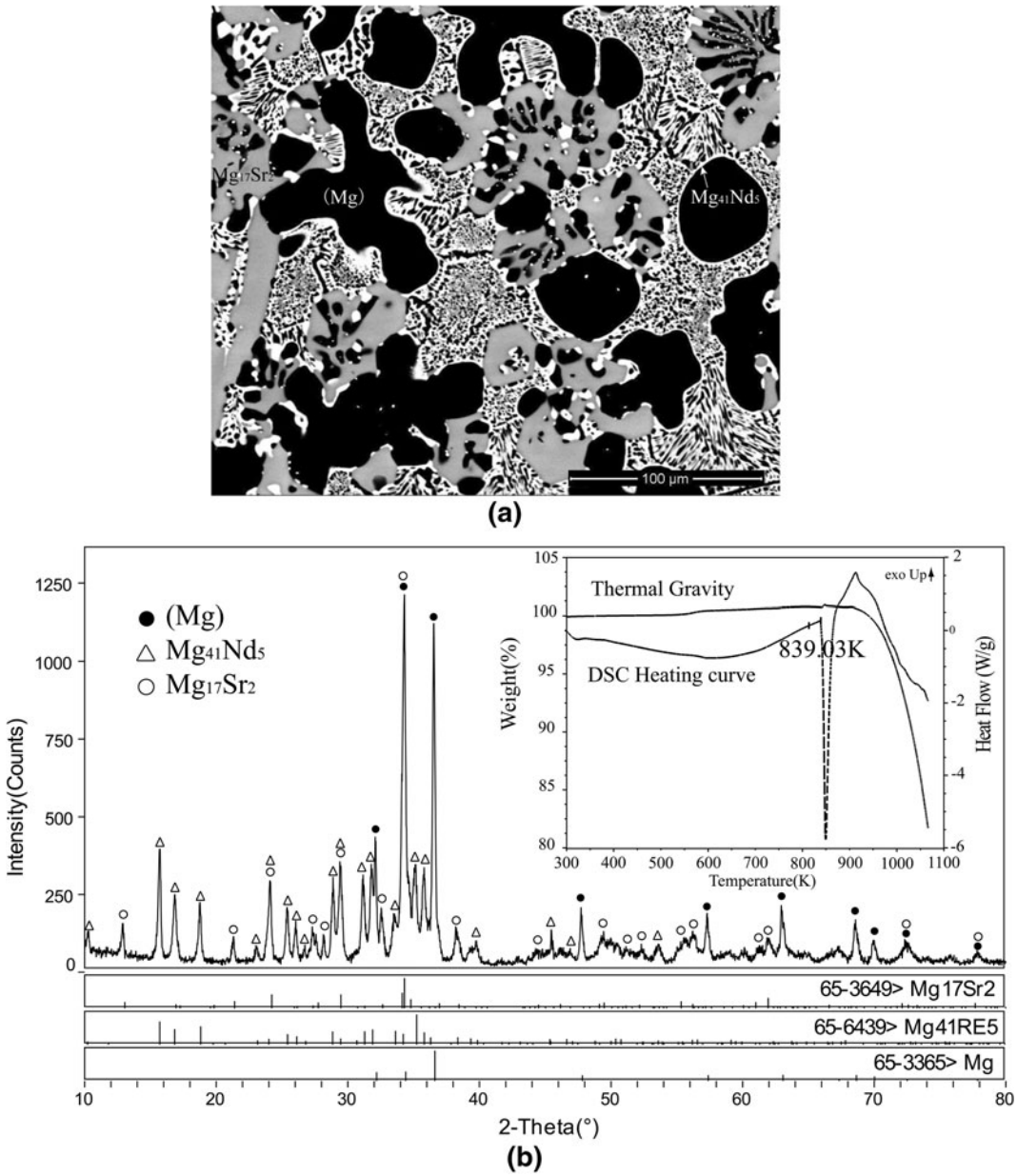

Fig. 1—(*a*) SEM back-scattered electron image, (*b*) XRD pattern and DSC heating curve of the Mg-2.52 at. pct Nd-3.31 at. pct Sr (#1) alloy after being annealed at 673 K (400 °C) for 720 h.

larger microstructure than what the ternary eutectics exhibits, it thus should occur first. Such a solidification pathway would have to be used and represented by the following thermodynamic modeling. The DSC curve of Alloy #1 in Figure 1(c) sets the temperature of the ternary eutectic reaction Liq $\rightarrow$ (Mg) + $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$ to be 839 K (566 °C). Note that the absence of endothermic peak associated with the liquidus temperature is attributed to the volatilization of Mg at the temperatures that approach its melting temperature, which was demonstrated by our thermal gravity examination.

The microstructure observation to Alloy #3 (Mg-7.72 at. pct Nd-1.88 at. pct Sr) and Alloy #4 (Mg-11.10 at. pct Nd-2.07 at. pct Sr) allows two separate two-phase regions to be determined at the Mg-rich corner, one is $Mg_{41}Nd_5$ + (Mg) in view of the block-like $Mg_{41}Nd_5$ crystal and the [(Mg) + $Mg_{41}Nd_5$] eutectics from Figure 2(b) and the other is Mg_3Nd + $Mg_{41}Nd_5$ upon the Mg_3Nd primary and the $Mg_{41}Nd_5$ precipitate in Figure 2(c). The $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$ phase region was recognized by identifying the coexistence of the dark $Mg_{17}Sr_2$ matrix and the gray $Mg_{41}Nd_5$ precipitates mainly along the grain boundaries from the SEM image in Figure 3(a), the XRD pattern in Figure 3(b) and the EMPA data of Alloy #5 (Mg-2.75 at. pct Nd-7.92 at. pct Sr). The DSC curve in Figure 3(b) confirms that this alloy starts to melt at 860 K (587 °C) higher than the melting temperature 839 K (566 °C) of Alloy

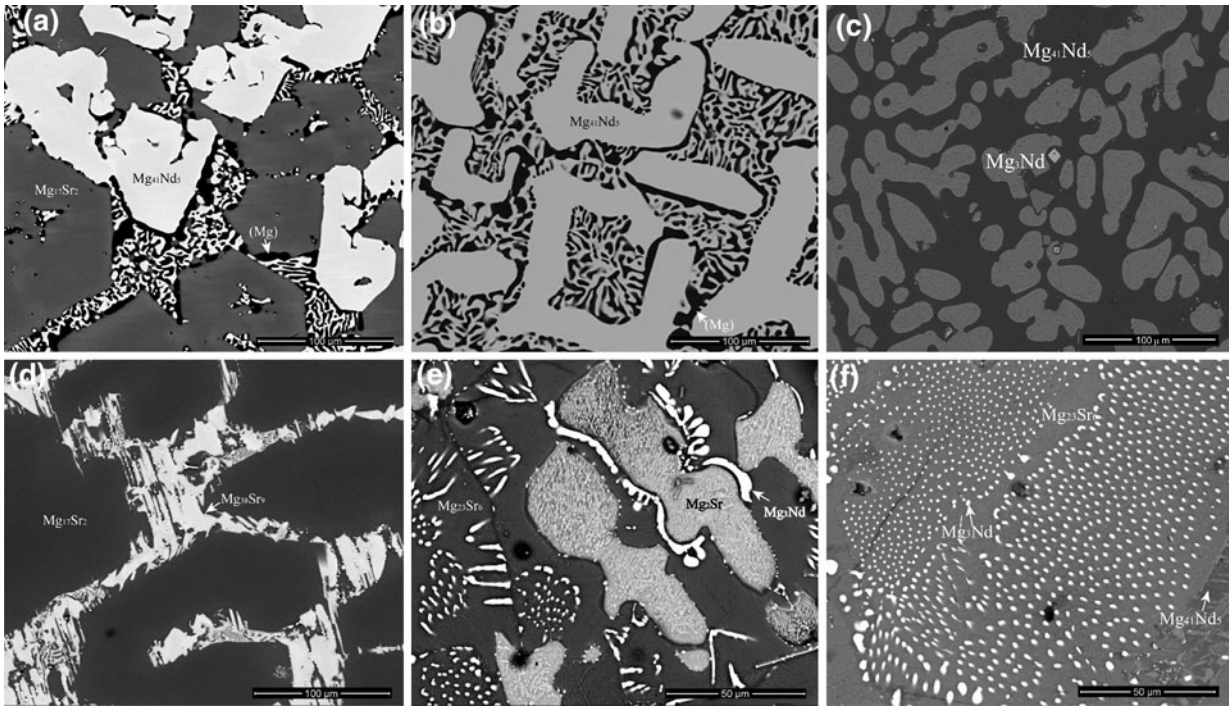

Fig. 2—The microstructure of (*a*) Mg-2.1 at. pct Nd-7.57 at. pct Sr (#2), (*b*) Mg-7.72 at. pct Nd-1.88 at. pct Sr (#3), (*c*) Mg-11.10 at. pct Nd-2.07 at. pct Sr (#4), (*d*) Mg-0.49 at. pct Nd-12.85 at. pct Sr (#6), (*e*) Mg-4.10 at. pct Nd-24.52 at. pct Sr (#7), and (*f*) Mg-4.51 at. pct Nd-12.23 at. pct Sr (#8) by SEM back-scattered electron observation.

#2 and 846 K (573 °C) of Alloy #6 (Mg-0.49 at. pct Nd-12.58 at. pct Sr), therefore implying a ridge-like shape of the $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$ two-phase region lifted by two three-phase equilibria from both sides. Note that no ternary compound has been discovered throughout the present experimentation.

Our experimental findings over eight ternary alloys, together with the phase diagrams of the Mg-Nd and Mg-Sr binaries, enable an isothermal section of the Mg-Nd-Sr ternary system at 673 K (400 °C) to be outlined and a liquidus projection to be estimated both at the Mg-rich corner, however, it is not yet a thorough study that supplies a panoramic picture of phase diagram for the ternary system. This can be taken from a critical thermodynamic modeling of the Mg-Nd-Sr ternary system by using the CALPHAD approach when the new experimental results are considered and represented in the modeling. In the context of the CALPHAD method, the solution phases, *i.e.* liquid, FCC_A1, BCC_A2, HCP_A3, and DHCP, are described by using the substitutional solution model, yielding the following expression for the Gibbs energy:

$$G^{\phi} = \sum_i x_i G_i^{\circ,\phi} + RT\sum_i x_i \ln x_i + \sum_{i,j>i} x_i x_j \sum_v L_{i,j}^{v,\phi}(x_i - x_j)^v, \quad [1]$$

where x_i and x_j represent the mole fractions of elements (i, j = Mg, Nd, Sr); $G^{\circ}$is the Gibbs energy of pure element i with a reference of standard state [298 K (25 °C) and 1 bar]; $L_{i,j}^{v,\phi}$ stands for the vth order interaction parameter. The two sub-lattices model was adopted to model all the compounds in the ternary system. Depending on the homogeneity range of the compounds, either $Mg_a(Nd,Sr)_b$ or $(Mg,Nd,Sr)_a(Mg,Nd,Sr)_b$ is used. The model yields the Gibbs energy as the following form:

$$G^{\phi} = \sum_i \sum_j y_i^I y_j^{II} G_{i:j}^{\circ,\phi} + RT\left(a\sum_i y_i^I \ln y_i^I + b\sum_j y_j^{II} \ln y_j^{II}\right) + {}^{ex}G^{\phi} \quad [2]$$

$$\begin{aligned} {}^{ex}G^{\phi} = & \sum_m y_m^{II} \sum_{i,j>i} y_i^I y_j^I \sum_v L_{i,j}^{v,\phi,I}\left(y_i^I - y_j^I\right)^v \\ & + \sum_i y_i^I \sum_{m,n>m} y_m^{II} y_n^{II} \sum_v L_{m,n}^{v,\phi,II}\left(y_m^{II} - y_n^{II}\right)^v \\ & + \sum_{i,j} \sum_{m,n} y_i^I y_j^I y_m^{II} y_n^{II} L^{rec}, \end{aligned} \quad [3]$$

where y_i^I is the site fraction of i element in I sublattice, $G_{i:j}^{\circ,\phi}$ is the Gibbs energies of the end-member compounds, and L^{rec} represents the reciprocal parameter.

The thermodynamic description of the Mg-Nd binary was taken from Qi *et al.*[11] with a minor modification and Mg-Sr from Zhong *et al.*[6] Due the lack of a thermodynamic description for the Nd-Sr binary, the present thermodynamic modeling started from assessing the Gibbs energy of the Nd-Sr binary phases by fitting to the phase diagram data from Reference 12. Upon the assessed thermodynamic parameters for Nd-Sr in Table -

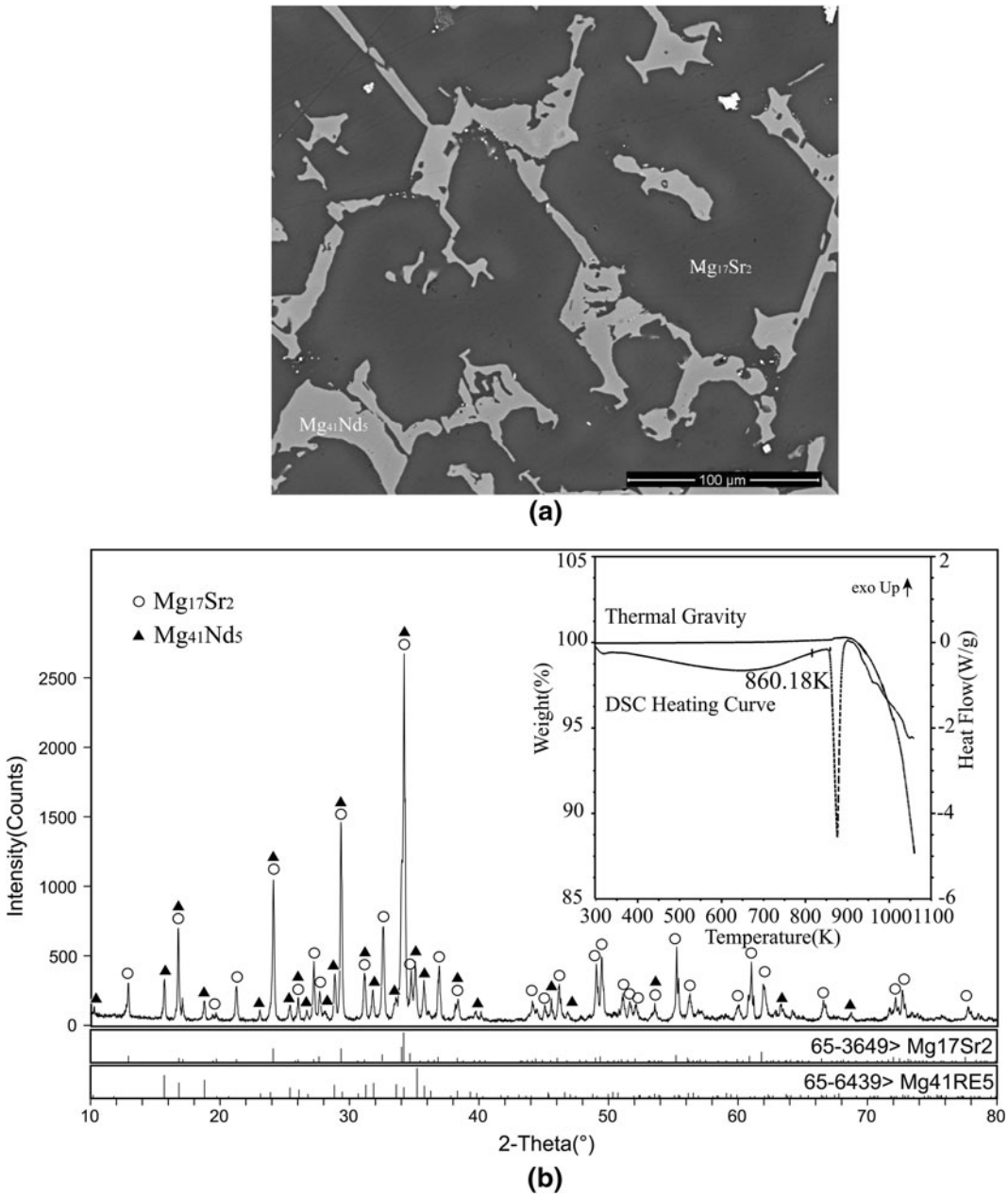

Fig. 3—(*a*) SEM back-scattered electron image, (*b*) XRD pattern and DSC heating curve of the Mg-2.75 at. pct Nd-7.92 at. pct Sr (#5) alloy after being annealed at 673 K (400 °C) for 720 h.

II, the calculated Nd-Sr phase diagram in Figure 4 well represents the reported invariant reactions and the solidus and liquidus temperatures, particularly for the solidus and liquidus at the Nd rich side. The reaction Liq → DHCP + BCC_A2 is predicted to occur at 1048 K (775 °C) rather than 1013 K (740 °C) in order to avoid an "improper" deep eutectic valley[13] lying extremely close to pure Sr in view of an updated melting temperature of 1050 K (777 °C) for Sr.[14] Note that there is no thermochemical data of Nd-Sr available for the assessment and comparison. This is not crucial to the Nd-Sr binary owing to the absence of intermetallic compound. However, an assessment without thermochemical data is not reliable for the Mg-Nd-Sr ternary.

As a consequence, the formation enthalpy hypothetical end-member compounds, *i.e.* $Mg_{41}Sr_5$, $Mg_{17}Nd_2$, $Mg_{38}Nd_9$ and $Mg_{23}Nd_6$, were predicted by physically sound first-principles calculation, see the values in italic of Table II. The calculation was performed in Vienna *ab initio* simulation package (VASP)[15] with the projector augmented wave pseudopotential formalism[16] and the generalized gradient approximation[17] for exchange correlation energy. A constant plane-wave energy cutoff of 350 eV was set in the calculations. Brillouin zone integrations were performed using Monkhorst–Pack *k*-point meshes.[18] A smearing parameter of 0.1 eV was adopted. The numerical converging threshold for formation enthalpy was set to be 1×10^{-6} eV/unit. The

Table II. Assessed Thermodynamic Parameters for the Mg-Nd-Sr System

Phase	Thermodynamic Modeling	Parameters		Reference
Liquid	(Mg,Nd,Sr)	${}^{0}L_{Mg,Nd}$	$-34{,}308.7 + 10.9887*T$	[11]
		${}^{1}L_{Mg,Nd}$	$-14{,}565.6$	[11]
		${}^{2}L_{Mg,Nd}$	$-16{,}479$	[11]
		${}^{0}L_{Mg,Sr}$	$-20{,}858 + 6.358*T$	[6]
		${}^{1}L_{Mg,Sr}$	$-13{,}008 + 4.715*T$	[6]
		${}^{0}L_{Nd,Sr}$	$+38{,}003.8$	this work
		${}^{1}L_{Nd,Sr}$	$-24{,}009.6 + 7.4*T$	this work
		${}^{2}L_{Nd,Sr}$	3000	this work
BCC_A2	(Mg,Nd,Sr):Va	${}^{0}L_{Mg,Nd:Va}$	$-24{,}721.4 + 1.5046*T$	[11]
		${}^{1}L_{Mg,Nd:Va}$	$-25{,}696.4 + 8.2963*T$	[11]
		${}^{2}L_{Mg,Nd:Va}$	$-16{,}898.5$	[11]
		${}^{0}L_{Mg,Sr:Va}$	20,000	[6]
		${}^{0}L_{Nd,Sr:Va}$	45,013.5	this work
		${}^{1}L_{Nd,Sr:Va}$	$-20{,}000$	this work
		${}^{2}L_{Nd,Sr:Va}$	5000	this work
HCP_A3-(Mg)	(Mg,Nd,Sr):Va	${}^{0}L_{Mg:Nd:Va}$	$-18{,}000$	[11]
		${}^{1}L_{Mg:Nd:Va}$	-7000	[11]
		${}^{0}L_{Mg:Sr:Va}$	10,000	[6]
		${}^{0}L_{Nd,Sr:Va}$	49,906.65	this work
DHCP	(Mg,Nd,Sr)	${}^{0}L_{Mg,\ Nd}$	$22{,}713-5.6252*T$	[11]
		${}^{1}L_{Mg,Nd}$	38,913.4	[11]
		${}^{0}L_{Nd,Sr}$	24,514.7	this work
		${}^{1}L_{Nd,Sr}$	-1000	this work
FCC_A1	$(Mg,Nd,Sr)_1$:Va_3	${}^{0}L_{Mg,Sr:Va}$	20,000	[6]
		${}^{0}L_{Nd,Sr:Va}$	$80{,}063.26 + 7.0102*T$	this work
$Mg_{41}Nd_5$	$(Mg,Nd)_{41}(Mg,Nd,Sr)_5$	$G_{Mg:Mg}$	$+138{,}000 + 46*G_{Mg}^{\circ}$	[11]
		$G_{Mg:Nd}$	$-459{,}385.6 + 217.3474*T + 5*G_{Nd}^{\circ} + 41*G_{Mg}^{\circ}$	[11]
		$G_{Nd:Mg}$	$+459{,}385.6-217.3474*T + 41*G_{Nd}^{\circ} + 5*G_{Mg}^{\circ}$	[11]
		$G_{Nd:Nd}$	$+138{,}000 + 46*G_{Nd}^{\circ}$	[11]
		$G_{Mg:Sr}$	$\mathit{-179{,}047.846} + 42.5*T + 41*G_{Mg}^{\circ} + 5*G_{Sr}^{\circ}$	this work
		${}^{0}L_{Mg,Nd:Mg}$ & ${}^{0}L_{Mg,Nd:Nd}$	$-51{,}679.8 + 10.1731*T$	[11]
		${}^{0}L_{Mg:Mg,Nd}$ & ${}^{0}L_{Nd:Mg,Nd}$	$-56{,}137.8 + 3.7333*T$	[11]
		${}^{0}L_{Mg:Nd,Sr}$	$-100{,}000$	this work
$Mg_{17}Sr_2$	$Mg_{17}(Nd,Sr)_2$	$G_{Mg:Sr}$	$-90{,}039 + 14.054*T + 17*G_{Mg}^{\circ} + 2*G_{Sr}^{\circ}$	[6]
		$G_{Mg:Nd}$	$\mathit{-115{,}165.47} + 91.91*T + 17*G_{Mg}^{\circ} + 2*G_{Nd}^{\circ}$	this work
		$G_{Mg:Nd,Sr}$	$-74{,}000$	this work
$Mg_{38}Sr_9$	$Mg_{38}(Nd,Sr)_9$	$G_{Mg:Sr}$	$-338{,}998 + 54.726*T + 38*G_{Mg}^{\circ} + 9*G_{Sr}^{\circ}$	[6]
		$G_{Mg:Nd}$	$\mathit{-373{,}982.29} + 115.21*T + 38*G_{Mg}^{\circ} + 9*G_{Nd}^{\circ}$	this work
		$G_{Mg:Nd,Sr}$	$-50{,}000$	this work
$Mg_{23}Sr_6$	$Mg_{23}(Nd,Sr)_6$	$G_{Mg:Sr}$	$-222{,}032 + 37.283*T + 23*G_{Mg}^{\circ} + 6*G_{Sr}^{\circ}$	[6]
		$G_{Mg:Nd}$	$\mathit{-260{,}683.268} + 108.43*T + 23*G_{Mg}^{\circ} + 6*G_{Nd}^{\circ}$	this work
		$G_{Mg:Nd,Sr}$	$-40{,}000$	this work
Mg_2Sr	$(Mg,Nd,Sr)_2(Mg,Nd,Sr)$	$G_{Mg:Mg}$	$+15{,}000 + 3*G_{Mg}^{\circ}$	[21]
		$G_{Sr:Mg}$	$+108{,}788 + 2*G_{Sr}^{\circ} + G_{Mg}^{\circ}$	[6]
		$G_{Mg:Nd}$	$-26{,}100 + 3*T + 2*G_{Mg}^{\circ} + G_{Nd}^{\circ}$	[11]
		$G_{Mg:Sr}$	$-29{,}661 + 3.031*T + 2*G_{Mg}^{\circ} + G_{Sr}^{\circ}$	[6]
		$G_{Sr:Sr}$	$+21{,}753 + 3*G_{Sr}^{\circ}$	[6]
		$G_{Nd:Mg}$	$+56{,}100 + 2*G_{Nd}^{\circ} + G_{Mg}^{\circ}$	[11]
		${}^{0}L_{Mg,Sr:Sr}$	$+35{,}000$	[6]
		${}^{0}L_{Mg:Nd,Sr}$	$+15{,}000$	this work

The values in italic are calculated by the first principle calculation.

calculated enthalpy values were adapted to the Redlich–Kister expression of the four end-members in the current thermodynamic assessment, therefore providing reliable solid-state energetics. This is particularly true for describing the $Mg_{41}Nd_5$ phase when the entropy term of its $Mg_{41}Sr_5$ end-member was further assessed by fully considering and representing the phase equilibria over two Mg-RE-Sr ternary systems (RE = Gd, Nd). Other ternary parameters were optimized by fitting to the present experimental data and are listed in Table II. Note that the $Mg_{12}Nd$ phase addressed in the literature[19,20] was not taken into consideration as there is no trace of the phase in our sample alloys. Thermodynamic modeling based on the assessed parameters allows good prediction of all phase equilibria and thermochemistry of the Mg-Nd-Sr system. Figure 5(a) shows that the

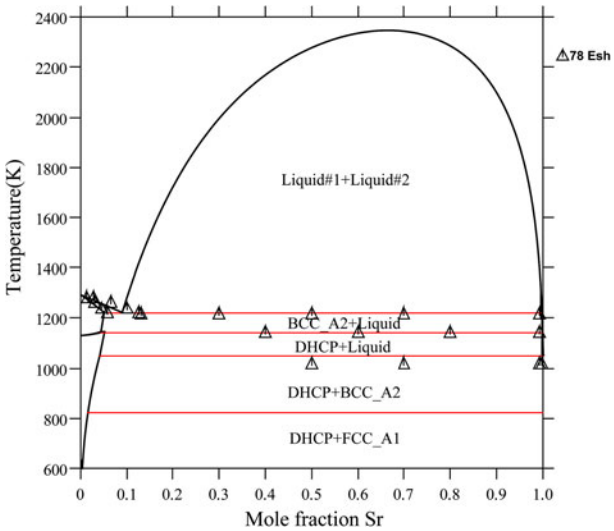

Fig. 4—Calculated phase diagram of the Nd-Sr binary system in comparison with data in Ref. [12].

calculated Mg-rich isothermal section at 673 K (400 °C) satisfactorily describes all the phase present and phase compositions determined by XRD and SEM/EPMA. The reasonable agreement of the calculated liquidus projection with the observed primary crystallization fields is illustrated in Figure 5(b) and of the calculated temperature of the Liq $\rightarrow$ (Mg) + $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$ reaction and the melting behavior of Alloy #5 with the measured DSC signals in the isopleth of Mg-2.52 at. pct Nd-3.31 at. pct Sr/Mg-2.75 at. pct Nd-7.92 at. pct Sr in Figure 5(c).

The thermodynamic modeling also offers an efficient way to study the solidification pathways. A representative solidification path for Alloy #1 is predicted and plotted in Figure 5(b) on the liquidus projection depicting the microstructure development from the (Mg) primary crystals to the coarse [(Mg) + $Mg_{17}Sr_2$] eutectics when the composition of liquid reaches the (Mg) + $Mg_{17}Sr_2$ mono-variant line and then to the fine [(Mg) + $Mg_{41}Nd_5$ + $Mg_{17}Sr_2$] ternary eutectics where complete solidification occurs.

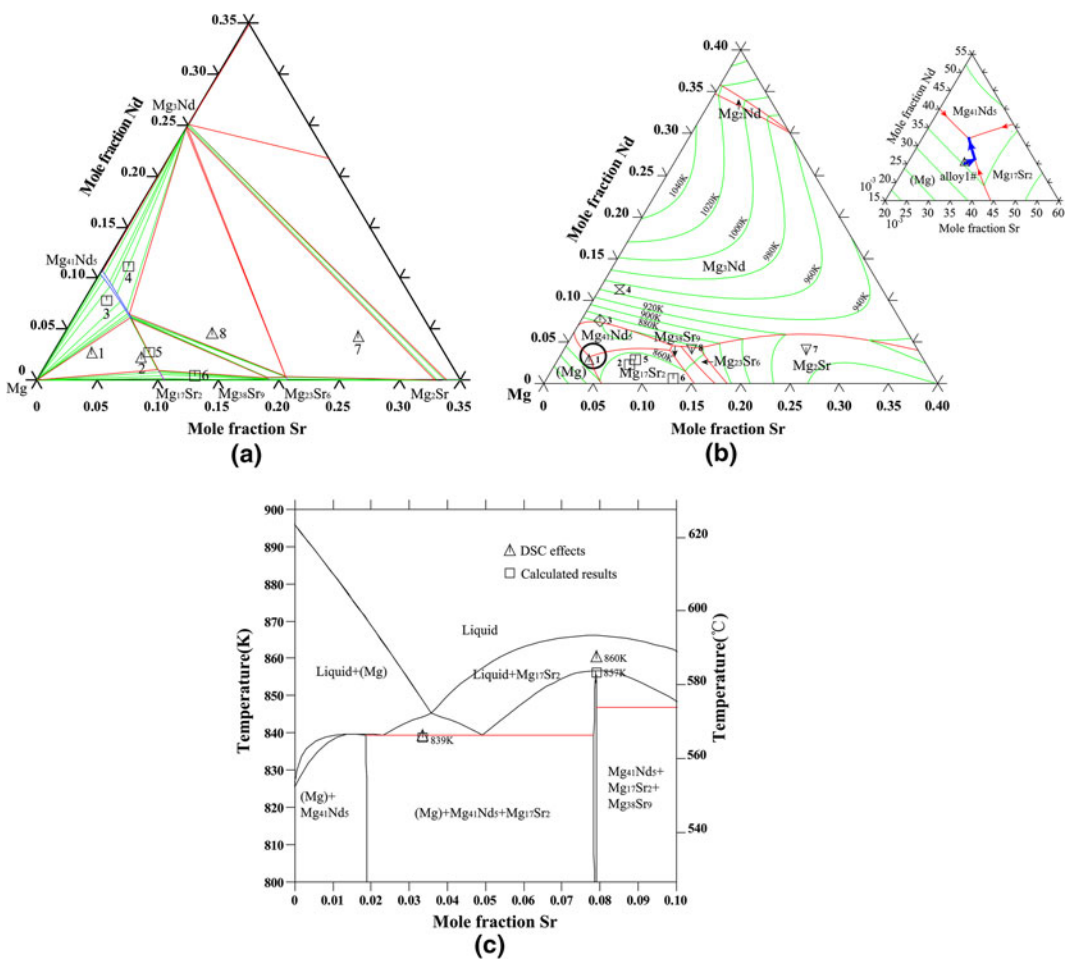

Fig. 5—(*a*) The calculated partial isothermal section at 673 K (400 °C) of the Mg-Nd-Sr system according to the present work. Symbols are the experimental data obtained from this work. It is the best interpretation of phases present and microstructural characterizations of 8 ternary alloys investigated in this work. (*b*) Calculated liquidus projection in comparison with the primary crystal observed from this experimental investigation and calculated diagram of solidification pathway for Mg-2.52 at. pct Nd-3.31 at. pct Sr (#1) alloy in magnified liquidus. (*c*) Calculated isopleth of Mg-2.52 at. pct Nd-3.31 at. pct Sr (#1)/Mg-2.75 at. pct Nd-7.92 at. pct Sr (#5).

In summary, the phase equilibria of the Mg-Nd-Sr ternary system at the Mg-rich corner was investigated by performing the microstructure examinations and phase determination on eight annealed alloys at 673 K (400 °C) and the DSC measurements. The experimental phase equilibria as-received, as well as those in the literature,[6,11,12] were assessed to develop the self-consistent thermodynamic descriptions of the Nd-Sr binary and Mg-Nd-Sr ternary in the framework of the CALPHAD method. The thermodynamic modeling based on the assessed parameters was shown capable of fully representing the isothermal section, the liquidus projection, the invariant reactions and the solidification pathway, therefore having potentials to offer a panoramic picture of phase diagram and thermochemistry for the Mg-Nd-Sr ternary system at the Mg-rich corner.

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